# CampusWatch x360 Energy Action Engine

"CampusWatch x360 is a proposed campus twin for energy, water, space/occupancy, mobility and food. Its Energy Action Engine uses existing meters, timetables and occupancy estimates to predict demand, flag anomalies, rank actions by savings/confidence/urgency/effort, compare feasible scenarios, verify approved actions against adjusted baselines, log outcomes and guide budgets. Explain this workflow and recommend low-cost methods with citations. Occupancy forecasts do not confirm live presence." *Generated by LeapSpace*

## CampusWatch x360 Energy Action Engine at a Glance

### Key Findings Table

| Topic | Key finding | Relevant population / setting | Citations |
|---|---|---|---|
| **Overall workflow** | A practical Energy Action Engine can be built as a meter-first pipeline: ingest context, forecast demand, flag anomalies, generate and rank actions, compare scenarios, verify savings against adjusted baselines, then log outcomes for future budgeting. | Campus and university buildings; operational energy management | [1] [2] [3] [4] [5] [6] [7] [8] |
| **Forecasting inputs** | Existing meters become more useful when paired with weather, academic calendars, timetables, and occupancy proxies rather than used alone. | University campuses and academic buildings | [2] [9] [10] [11] |
| **Forecasting models** | Low-cost regression or change-point baselines are operationally attractive; SARIMAX, XGBoost, GPR, and hybrids can improve accuracy when enough history exists. | Campus electricity and building load forecasting | [2] [9] [12] |

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| **Occupancy role** | Occupancy improves forecasting, calibration, and control, but timetable, Wi-Fi, CO2, and booking signals are probabilistic proxies, not confirmation of live presence. | University rooms, buildings, and campuses | [9] [13] [14] [15] [16] [17] |
| **Anomaly detection** | Low-cost anomaly screening works best with interval-meter profiles, clustering, change-point detection, residual thresholds, and rule-based checks on existing BMS points. | Campus/public/commercial building energy data | [3] [18] [19] [20] |
| **False-positive control** | Weather and operating-regime effects must be separated from true faults using baseline models, regime proxies, and continuously updated normal patterns. | Building energy anomaly and FDD workflows | [21] [22] [23] [24] |
| **Action ranking** | There is no retrieved standard campus method that already ranks actions by the exact quartet of savings, confidence, urgency, and effort, but MCDA methods are adaptable. | Energy planning, retrofit prioritization, campus decision support | [4] [5] [12] [25] [26] [27] |
| **Scenario comparison** | Digital twins can compare lighting, HVAC, setpoint, occupancy-control, preheating, and retrofit alternatives before approval using energy, cost, comfort, and carbon metrics. | University and building twins | [28] [29] [30] [31] [32] |
| **Savings verification** | Credible post-action savings require adjusted baselines rather than raw pre/post comparison; weather, operating hours, occupancy shifts, shutdowns, and recommissioning matter. | Building and campus M&V | [6] [7] [8] [33] [34] [35] |

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| **Budget support** | Logging anomalies, actions, baseline changes, verified savings, and uncertainty creates institutional memory for phased budgets and portfolio planning. | Campus operations and capital planning | [8] [36] [37] [38] |

### Direct Answer

CampusWatch x360's Energy Action Engine is feasible as a low-cost, meter-first decision workflow. The strongest evidence supports this sequence: use existing interval meters and BMS points, add weather plus academic-calendar and timetable context, derive probabilistic occupancy features, forecast demand, detect abnormal deviations from expected use, generate candidate operational actions, rank them with a transparent multi-criteria score, compare feasible scenarios, approve selected actions, verify results against adjusted baselines, and log outcomes to improve later budget choices [1] [2] [3] [4] [5] [6] [7] [8].

For low-cost implementation, the best-supported starting stack is:

1. **Forecasting:** regression or change-point baselines using meter data, outdoor temperature, day type, semester/exam/holiday flags, and timetable-derived occupancy proxies; then benchmark SARIMAX, GPR, XGBoost, or a hybrid only if extra complexity pays off [2] [9] [10] [11] [12].
2. **Occupancy inputs:** use timetables and calendars first; optionally add aggregated Wi-Fi AP counts, booking totals, or badge totals if governance allows; treat all of them as uncertain occupancy estimates rather than proof of live presence [9] [14] [15] [16] [39].
3. **Anomaly detection:** use weekly profile clustering, residual thresholds, change-point detection, and rule-based BMS checks; isolation forest is a useful second-stage screen when labeled faults are unavailable [3] [18] [19] [20].
4. **Action ranking:** use a simple weighted score or MCDA method such as AHP or PROMETHEE; a practical design is `Score = w1*savings + w2*confidence + w3*urgency – w4*effort`, followed by sensitivity testing and then a separate budget constraint step [4] [5] [38].
5. **M&V:** verify approved actions against adjusted baselines that account for weather, operating hours, and non-routine events; keep versioned models and event logs because model choice changes reported savings [7] [8] [34] [35].

**Important constraint:** occupancy forecasts can improve planning and control support, but they do **not** confirm that a space is occupied right now [14] [16] [17].

**Confidence level: High** - The workflow is strongly supported in pieces across forecasting, anomaly screening, MCDA, and M&V literatures, though the exact all-in-one CampusWatch formula is an informed synthesis rather than a directly published standard [1] [3] [6] [25].

### Study Scope

This report synthesizes evidence from campus/building digital twins, university energy forecasting, occupancy estimation, anomaly detection/FDD, and measurement-and-verification studies described in the supplied record, with emphasis on low-cost methods using existing infrastructure and on university or comparable public/commercial buildings [1] [3] [6] [12] [14]. Methods covered include statistical forecasting, machine learning, hybrid models, rule-based diagnostics, unsupervised anomaly screening, MCDA, and baseline-adjusted M&V [2] [5] [19] [35] [40].

### Assumptions & Limitations

The supplied evidence supports each workflow component well, but not a single published campus twin that already integrates the exact ranking dimensions of savings, confidence, urgency, and effort in one validated engine [25] [27]. Evidence is also stronger for electricity/HVAC workflows than for a fully integrated campus twin spanning energy, water, space, mobility, and food [6] [12]. Timetable-derived occupancy appears useful, but the retrieved studies rarely isolate its marginal forecast contribution with clean ablation tests [2] [9] [41].

### Suggested Further Research

- Quantify the incremental value of timetable-only versus Wi-Fi-augmented occupancy features on forecast error and anomaly quality [41] [42].
- Test end-to-end campus workflows

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